



## NORTHERN WOLVES AIR CONDITIONING

# Request for Information

## RFI #01 - REVISION 1 - PROPOSED SOLUTION

|                                                                     |                                          |
|---------------------------------------------------------------------|------------------------------------------|
| <b>PROJECT:</b> BOSS BELMONT PARK                                   | <b>CM:</b> JRMC                          |
| <b>ENGINEER:</b> PETER A                                            | <b>ARCHITECT:</b> KENNETH PARK           |
| <b>NORTHERN WOLVES RFI#:</b> 01 - Rev. 1 (Response to EOR comments) | <b>DATE:</b> 09-11-2026                  |
| <b>SUBJECT:</b> VRF BRANCH BOX EXHAUST VENT                         | <b>DRAWING NO:</b>                       |
| <b>ORIGINAL RFI DATE:</b> 08-20-2026                                | <b>EOR RESPONSE RECEIVED:</b> 08-20-2026 |

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### Reference documents

Mitsubishi Electric Application Note 2045, "New X-generation Ventilated Branch Circuit (BC) Controller," May 2026.

Mitsubishi Electric Installation Manual WT11269X01, CMB-M-NU-MA(B)1-SV, Sections 5.7 and 6.

**Submitted by:** Northern Wolves Air Conditioning

**Distribution:** Peter A (Engineer), Kenneth Park (Architect), JRMC (CM)

## RFI #01 - VRF Branch Box Exhaust Vent

### PROPOSED SOLUTION

In response to the Engineer's comments dated 08-20-2026:

#### 1. ERV connection withdrawn

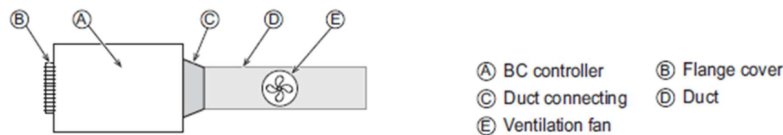
Northern Wolves withdraws the proposal to connect the branch box to the ERV exhaust. The branch box will be ventilated only on a refrigerant leak event, as required by the Engineer and by Mitsubishi.

#### 2. Ventilated branch box confirmed

Mitsubishi CITY MULTI **CMB-M1012NU-MA1-SV** (ventilated 12-branch BC controller) is furnished for this project. It has a built-in refrigerant leak sensor, a 4" duct connection port and a leak-detection output signal. It has no fan. The fan, duct and relay are field-provided.

#### 3. Proposed solution: dedicated leak-activated exhaust fan, 4" duct to outdoors

|                     |                                                                                                                                                                                                                                                                                                                                                                    |
|---------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Exhaust fan</b>  | <b>Fantech FG 4</b> inline centrifugal fan, 4" duct connection, 120 V / 1 ph, 0.19 A, 20 W, UL listed, galvanized steel housing. 135 CFM free air, 110 CFM at 0.2 in. w.c. One fan per BC controller. See Attachment A.                                                                                                                                            |
| <b>Design basis</b> | Mitsubishi App Note 2045: 88 CFM minimum, 588 CFM maximum, 0.08 in. w.c. negative pressure in enclosure, 0.12 in. w.c. loss through unit, fan kVA of 1.0 or less, no ignition source in airstream. The selected fan meets all criteria.                                                                                                                            |
| <b>Ductwork</b>     | 4" round galvanized duct from one BC controller side port (opposite port stays factory-capped) to the exterior wall. Joints sealed, first 39" insulated, supported from structure, no sharp bends. Total run limited to 20 equivalent feet including fittings; where the run is longer, Fantech FG 4XL (same 4" connection, 134 CFM at 0.4 in. w.c.) will be used. |
| <b>Discharge</b>    | To outdoors through an exterior wall hood with backdraft damper and bird screen. Duct sloped 1:30 down toward the outside. Hood located to maintain IMC 401.4 separation from outdoor air intakes and openings.                                                                                                                                                    |
| <b>Control</b>      | Fan runs only on refrigerant leak detection. BC controller output TBX1-2 energizes a field-provided relay; the relay switches a dedicated 120 V circuit to the fan. The VRF system shuts down on a leak. BC controller alarm contact TBX3-4 is available for BMS or local alarm if required by the Engineer.                                                       |



One BC controller to one ventilation fan (Mitsubishi App Note 2045, Fig. 7).

#### 4. Sketch

Northern Wolves' office will issue a sketch showing the duct routing, fan location, exterior discharge point and control wiring for the Engineer's review.

#### 5. Requested from the Engineer

Approve the proposed solution. Confirm the exterior discharge location with the Architect. Dedicated 120 V circuit to the fan relay by the Electrical Contractor.

# FG Series

## Inline Centrifugal Fans

The company that invented the circular duct fan more than thirty years ago, now introduces new improved inline fans that are virtually airtight.

New production processes eliminate manufacturing variances to ensure tight tolerances for optimal performance of every fan.

Galvanized steel housings meet code requirements for both residential and commercial applications. High quality ebm-papst motors ensure long life and dependable performance.

### Simplified Installation

The FG Series features new longer duct connections for easier installation. Fans are prewired and include bracket for mounting to wall or ceiling.

Straight-through airflow allows fans to be installed at any angle at any point along the ductwork.

### We've set a new standard for:

- Performance
- Efficiency
- Reliability
- Air Tightness
- Ease of Installation

### Features and Benefits

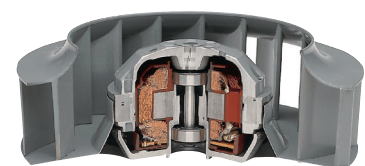
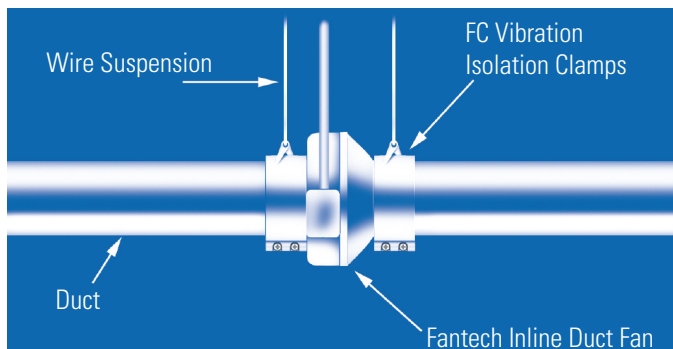
- 135 to 940 CFM
- 4" to 12" (102 to 305 mm) duct diameter
- Airtight galvanized steel housing – perfect for commercial code applications
- Powered by ebm-PAPST external rotor motorized impeller for excellent heat dissipation, even at low RPM
- Permanently lubricated sealed ball bearings for maintenance-free operation
- Automatic reset thermal overload protection
- Suitable for airstream temperatures up to 140° F (60° C)
- 100% speed controllable
- Rated for outdoor use
- UL Listed
- 5 year warranty



Fantech, Inc. certifies that the models shown herein are licensed to bear the AMCA Seal. The ratings are based on tests and procedures performed in accordance with AMCA Publication 211 and comply with the requirements of the AMCA Certified Ratings Program.



### Applications



Also Available: FGC Series  
Same fans prewired with six foot cord and plug

## Specification Data

| Model   |  | Rated power | Voltage / phase | Max amps | RPM               | 0.0" P <sub>s</sub> | 0.2" P <sub>s</sub> | 0.4" P <sub>s</sub> | 0.6" P <sub>s</sub> | 0.8" P <sub>s</sub> | 1.0" P <sub>s</sub> | 1.5" P <sub>s</sub> | Max P <sub>s</sub> | Shipping weight |    |
|---------|-----------------------------------------------------------------------------------|-------------|-----------------|----------|-------------------|---------------------|---------------------|---------------------|---------------------|---------------------|---------------------|---------------------|--------------------|-----------------|----|
|         |                                                                                   | W           | V / ~           | A        | min <sup>-1</sup> | cfm                 |                     |                     |                     |                     |                     |                     | in.wg              | lbs             | kg |
| FG 4    | •                                                                                 | 20          | 120/1           | 0.19     | 3000              | 135                 | 110                 | 83                  | 55                  | 25                  | -                   | -                   | 0.94               | 7               | 3  |
| FG 4XL  |                                                                                   | 71          | 120/1           | 0.66     | 2700              | 170                 | 150                 | 134                 | 119                 | 103                 | 86                  | 40                  | 1.98               | 8               | 4  |
| FG 5    | •                                                                                 | 20          | 120/1           | 0.19     | 3000              | 156                 | 130                 | 99                  | 66                  | 33                  | -                   | -                   | 0.99               | 7               | 3  |
| FG 5XL  |                                                                                   | 73          | 120/1           | 0.68     | 2700              | 220                 | 190                 | 160                 | 135                 | 112                 | 91                  | 41                  | 1.89               | 8               | 4  |
| FG 6    | •                                                                                 | 72          | 120/1           | 0.58     | 2700              | 303                 | 287                 | 270                 | 232                 | 196                 | 164                 | 58                  | 1.88               | 10              | 5  |
| FG 6M   |                                                                                   | 120         | 120/1           | 1.02     | 2350              | 418                 | 370                 | 317                 | 268                 | 224                 | 186                 | 101                 | 2.10               | 12              | 5  |
| FG 6XL  |                                                                                   | 153         | 120/1           | 1.48     | 2900              | 483                 | 450                 | 409                 | 369                 | 329                 | 289                 | 201                 | 2.41               | 12              | 5  |
| FG 8    |                                                                                   | 119         | 120/1           | 1.14     | 2550              | 461                 | 410                 | 351                 | 295                 | 243                 | 191                 | 97                  | 2.11               | 12              | 5  |
| FG 8XL  |                                                                                   | 142         | 120/1           | 1.45     | 2950              | 502                 | 470                 | 428                 | 388                 | 351                 | 313                 | 218                 | 2.40               | 13              | 6  |
| FG 10   |                                                                                   | 138         | 120/1           | 1.43     | 3000              | 513                 | 480                 | 444                 | 407                 | 366                 | 324                 | 216                 | 2.36               | 12              | 5  |
| FG 10XL |                                                                                   | 196         | 120/1           | 1.96     | 3100              | 589                 | 560                 | 531                 | 503                 | 472                 | 441                 | 355                 | 3.02               | 14              | 6  |
| FG 12   |                                                                                   | 181         | 120/1           | 1.87     | 2600              | 741                 | 680                 | 601                 | 515                 | 434                 | 363                 | 236                 | 2.99               | 18              | 7  |
| FG 12XL |                                                                                   | 301         | 120/1           | 3.01     | 2900              | 940                 | 880                 | 819                 | 746                 | 670                 | 596                 | 425                 | 2.74               | 21              | 10 |

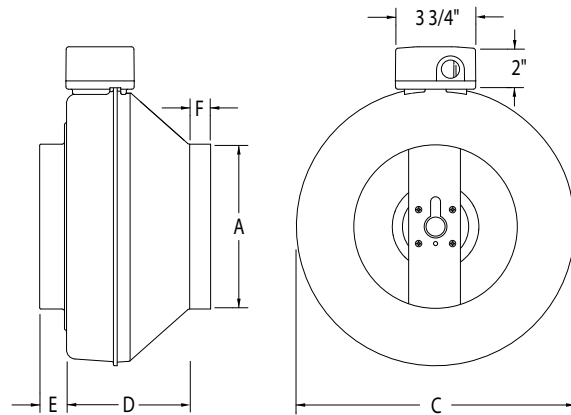
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Performance shown is for installation type D - Duct inlet, Ducted outlet. Speed (RPM) shown is nominal. Performance is based on actual speed of test. Performance ratings do not include the effects of appurtenances (accessories).

## Dimensions

| Model          | A        | C            | D             | E          | F          |
|----------------|----------|--------------|---------------|------------|------------|
| FG 4           | 4 (102)  | 8 1/2 (216)  | 6 1/2 (165)   | 1 (25)     | 1 (25)     |
| FG 4XL         | 4 (102)  | 9 3/4 (248)  | 6 15/16 (176) | 1 (25)     | 1 (25)     |
| FG 5           | 5 (127)  | 8 5/8 (219)  | 6 1/2 (165)   | 1 (25)     | 1 (25)     |
| FG 5XL         | 5 (127)  | 9 3/4 (248)  | 6 (152)       | 1 1/8 (29) | 1 1/8 (29) |
| FG 6           | 6 (152)  | 11 3/8 (289) | 6 1/4 (159)   | 1 (25)     | 7/8 (22)   |
| FG 6XL / FG 6M | 6 (152)  | 13 1/8 (333) | 7 (178)       | 1 (25)     | 1 (25)     |
| FG 8           | 8 (203)  | 13 1/4 (337) | 6 (152)       | 1 (25)     | 1 (25)     |
| FG 8XL         | 8 (203)  | 13 1/4 (337) | 6 (152)       | 1 1/8 (29) | 1 (25)     |
| FG 10          | 10 (254) | 13 1/4 (337) | 4 3/4 (121)   | 1 1/8 (29) | 1 (25)     |
| FG 10XL        | 10 (254) | 13 1/4 (337) | 4 13/16 (122) | 1 1/4 (32) | 1 (25)     |
| FG 12          | 12 (305) | 16 (406)     | 6 11/16 (170) | 1 1/4 (32) | 1 (25)     |
| FG 12XL        | 12 (305) | 16 (406)     | 6 11/16 (170) | 1 1/2 (38) | 1 (25)     |

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All dimensions in inches (mm).

†Duct connection is 1/8" smaller than duct size.

## Accessories



**FC**  
Mounting Clamps



**WC 15**  
Speed Control



**FLD 60**  
Light/Fan switch



**FD 60EM**  
Electronic Timer



**SCD**  
Speed Control



**FTD 7**  
7 Day Timer

## Customer Support:

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